San Diego Heat Risk Explorer User Guide

A primer on variables, methods, and map interpretation.

The Heat Risk Explorer is an interactive heat-risk map of San Diego County, computed at the census-tract level. To use the Heat Risk Explorer, users select variables from the available options (such as heat exposure, demographics, health conditions, air conditioning access, green space proximity, income, and others.) Once selected, the tool returns a composite **Heat Risk Index** and the dominant driver for each census tract.

ON THIS PAGE

1. Quick Start
2. Heat Risk Index Data Inputs
3. Reading the map
4. Examples
5. Feedback & bug reports
6. Data sources
7. Privacy

Quick Start

To get started, click **Open the Map** above.

1. Default variables are already selected — click **Build Composite Risk Map**.
2. Click any census tract to see its Heat Risk Index and dominant driver.
3. Under **Map Layer**, switch to **Dominant Driver** to see what drives risk for each census tract.
4. Or use the left sidebar to customize the map — toggle individual variables on or off before re-building.

Heat Risk Index Data Inputs

The Heat Risk Index is built from **three core pillars**: **heat exposure** (the physical heat experienced in a given census tract), **population sensitivity** (who lives there and the factors that make their communities more susceptible to heat), and **adaptive capacity** (systems and infrastructure that affect the ability of residents to cope with heat stress). Multiple variables (listed below) from each pillar can be selected by the user to be included.

Heat Exposure

Heat Exposure is **single-select**: you choose one of the four heat hazards below and the tool scores that hazard on its own, as the average of its underlying layers. The four are not combined, because they are physically distinct and affect different places — daytime and nighttime heat are close to spatially uncorrelated across San Diego County, so blending them would cancel out the pattern in both.

DAYTIME HEAT

Mean Daily Maximum Temperature; 85th and 99th Percentile Daily Maximum Temperature (the hottest 15% and 1% of days); and Annual Maximum Temperature. Peaks inland.

NIGHTTIME HEAT

Mean Overnight Minimum Temperature; 90th Percentile Overnight Minimum Temperature (warm nights); and No-Relief Night Fraction (the percent of nights without cooling relief). Peaks in dense urban and coastal census tracts. Warm nights are associated with elevated heat-illness risk, especially in older adults.

HUMID HEAT (WBGT)

85th and 99th Percentile Wet-Bulb Globe Temperature, which combines temperature, humidity, sun, and wind into a single heat-safety index. Humidity enters the tool here rather than as a standalone variable, because it modifies how dangerous a given air temperature is rather than acting as a hazard on its own.

SURFACE TEMPERATURE

Landsat-derived Mean Land Surface Temperature — the radiant urban-heat-island signal, measured at the ground rather than in the air.

Population Sensitivity

HEALTH

High Blood Pressure, Current Asthma, Coronary Heart Disease, COPD, Diabetes, Stroke, Depression, Cognitive Difficulty, and tract-level Life Expectancy percentile. Chronic-disease prevalence is from CDC PLACES; Life Expectancy is from the Healthy Places Index.

SOCIODEMOGRAPHICS

Age 17 & Under, Age 65+, Disability, Limited English, Single-Parent HH, Below 150% Poverty, Unemployed, Housing Cost Burden, No HS Diploma, Uninsured, Minority, Multi-Unit Structures, Mobile Homes, Crowding, No Vehicle, Group Quarters, and No Internet.

SOCIAL VULNERABILITY INDEX (SVI)

Pre-built SVI summary themes: Overall SVI, Socioeconomic Status, HH Disability/Language, Racial/Ethnic Minority, and Housing/Transportation. Source: CDC Social Vulnerability Index (SVI) 2022.

ENVIRONMENT

Pollution (Inverse), Ozone, PM 2.5, and Diesel PM exposure. Source: Healthy Places Index.

Higher population sensitivity raises the Heat Risk Index.

Adaptive Capacity

AIR CONDITIONING (AC)

AC Prevalence (the percent of homes with any AC) from two independent sources, listed in the sidebar as **AC Prevalence (LACE)** and **AC Prevalence (Romitti)**: the U.S. Census Bureau's Local Air Conditioning Estimates (LACE) 2023, the first official tract-level AC estimates from the Census Bureau, and Romitti et al. (2022), a published modeled estimate retained for comparison.

BUILT ENVIRONMENT

Tree Canopy, Park Access, Green Space (%), Renter-Occupied (%), and housing age (Pre-1980 and Pre-1960 Housing share). Sources: Healthy Places Index, ACS B25034, and SanGIS.

SOCIOECONOMIC

Income, Above Poverty, Economic, Education, Housing Quality, Healthcare Access, Transportation, Social Cohesion, Neighborhood Quality, Health Insurance, Automobile Access, Homeownership, Commute Time, Supermarket Access, Voter Participation, and Retail Food Access. Source: Healthy Places Index.

Higher adaptive capacity lowers the Heat Risk Index.

How the components combine

Each pillar score ranks a census tract relative to others in San Diego County. **Heat Exposure** is the average of the layers making up the heat mode you selected. **Population Sensitivity** and **Adaptive Capacity** are weighted combinations of the variables you turned on. All three are then percentile-ranked, and the Heat Risk Index is their mean, producing a single value between 0–100 per census tract.

Correlated variables are not double-counted. Lower-income tracts tend to have older housing, less tree cover, and higher chronic-illness rates. A direct sum would weight that shared pattern multiple times. Within Population Sensitivity and Adaptive Capacity, the tool uses Principal Component Analysis, a statistical method that identifies the dominant underlying patterns in a set of correlated variables, so each shared pattern is counted once rather than multiple times. Heat Exposure does not use PCA — a heat mode is a small set of readings of the same physical hazard, so its layers are simply averaged.

Adaptive-capacity variables enter the score with the opposite sign. A census tract with strong protective resources receives a lower Heat Risk Index even when its exposure is high.

Reading the map

Color

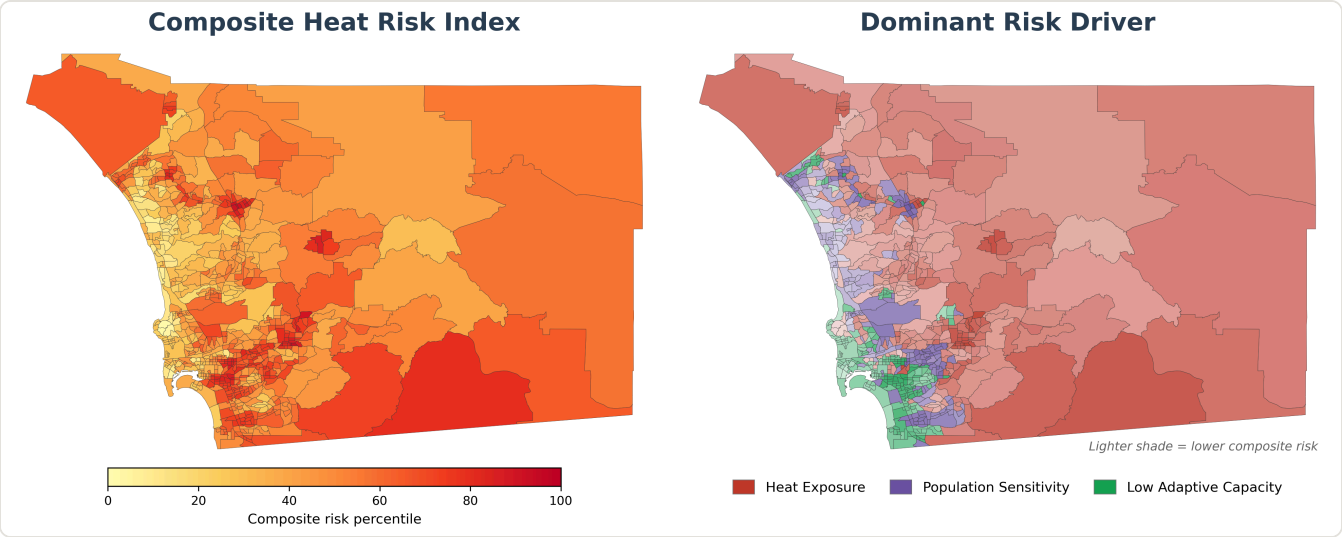
A darker color indicates a higher risk based on the composite Heat Risk Index. The composite map uses red (highest = darkest red). Single-pillar views use the pillar's color: red for Heat Exposure, purple for Population Sensitivity, green for Adaptive Capacity.

The **Dominant Driver** map reuses those three hues to mark which pillar drives each census tract's risk, and shade carries a second signal: **lighter = lower composite risk**. Note that green marks census tracts where *low* adaptive capacity is the leading driver — it flags a shortfall in cooling resources, not an abundance of them.

Turning on **Accessibility → colorblind-friendly palettes** replaces this scheme throughout. The composite ramp becomes viridis (purple through green to yellow), and the three drivers become orange (Heat Exposure), yellow (Population Sensitivity), and blue (low Adaptive Capacity).

Index interpretation

The Heat Risk Index is a *relative percentile*, not absolute risk. A Heat Risk Index of 75 means the census tract is at higher risk than 75% of San Diego County census tracts. It does not mean 75% of residents will be harmed.



San Diego County tracts shown two ways: the composite Heat Risk Index (left) and each tract's dominant risk driver (right).

Map layers

- **Composite Risk (left panel):** the 0–100 Heat Risk Index combining all three pillars (default view).

- **Dominant Driver (right panel):** census tracts are colored by which pillar contributes most to the Heat Risk Index. This distinguishes hotspots driven by Heat Exposure from those driven by Population Sensitivity or low Adaptive Capacity. Click on a census tract to see the scores for each of the three pillars.
- **Driver Mix:** a blend of all three drivers rather than only the winner. A census tract leans toward the hue of whichever driver it ranks highest on and reads white when the three are balanced; lighter still means lower composite risk. Use this where you want to spot census tracts with two comparable drivers, which the Dominant Driver map collapses to one.
- **Primary Spatial Patterns:** this drop-down menu lists the independent patterns identified by Principal Component Analysis across all selected variables. Each entry is labeled **Pattern N (var X%)**, where N is the pattern's rank by importance and X% is the share of variance in the data that this pattern explains. Hover over the blue information icons to see which variables contribute most to each pattern.

Census tract panel

Click a census tract for its three pillar scores, its Heat Risk Index, its dominant driver, and to compare it directly against other census tracts.

Sidebar panel

The left sidebar lets you customize the map. Sections, from top to bottom:

- **Heat Exposure:** a single-select list of the four heat hazards. Picking one replaces the exposure score rather than adding to it, so there is no header checkbox here.
- **Population Sensitivity / Adaptive Capacity:** checkbox groups. Use a pillar's header checkbox to toggle the whole pillar; expand each collapsible category (Health, Sociodemographics, Air Conditioning (AC), Built Environment, etc.) to toggle individual variables.
- **Build Composite Risk Map:** combines the variables you selected into a Heat Risk Index for every census tract.
- **Map Layer:** switch between the Composite Risk, Dominant Driver, a single pillar score, a single raw variable, or one of the Primary Spatial Patterns.
- **Jurisdictions:** overlay city or tribal-land boundaries on the map.
- **Climate Scenario:** change the temperature variables to projected mid- (2050) or end-of-century (2080) values under different emissions pathways. See below.
- **Accessibility:** switch to colorblind-friendly palettes.
- **Export:** save the current map as a PDF, or download the underlying census-tract-level data.

About the data download. *Download Data* gives you a CSV or GeoJSON of every tract, plus a second file — `<name>_data_dictionary.csv` — that explains it. Columns in the data file are raw variable keys such as `EP_NOINT` or `shututility_prev`; the dictionary gives each one a plain-English label, its pillar, its units, whether a higher value means *more* or *less* heat risk, and a link to the original source. Its header also records the heat mode, climate scenario and switched-on variables the index columns were computed from, so the download can be reproduced. GeoJSON files carry the same information in a `metadata` object. Two cautions worth reading before you analyse anything: `adaptive_score` is stored capacity-side (100 = most capacity, *least* risk), which is the opposite direction from `exposure_score` and `sensitivity_score`; and `_sd_pctile` columns are always ranked against all San Diego County tracts, even when you export a single city. A few variables appear on the map but are withheld from the download because their source's terms don't permit us to redistribute them: Healthy Places Index variables (non-commercial licence) and the CalHeatScore heat-hospitalisation variables (no published licence). The dictionary lists each of them with `in_download = no` and a note on where to request the data, so you can see exactly what is missing and why. Everything that *is* in the download is public-domain or openly licensed — see DATA_LICENSE.md for the full provenance audit.

Future scenarios

Three buttons at the top of the Climate Scenario panel switch the temperature variables between time periods:

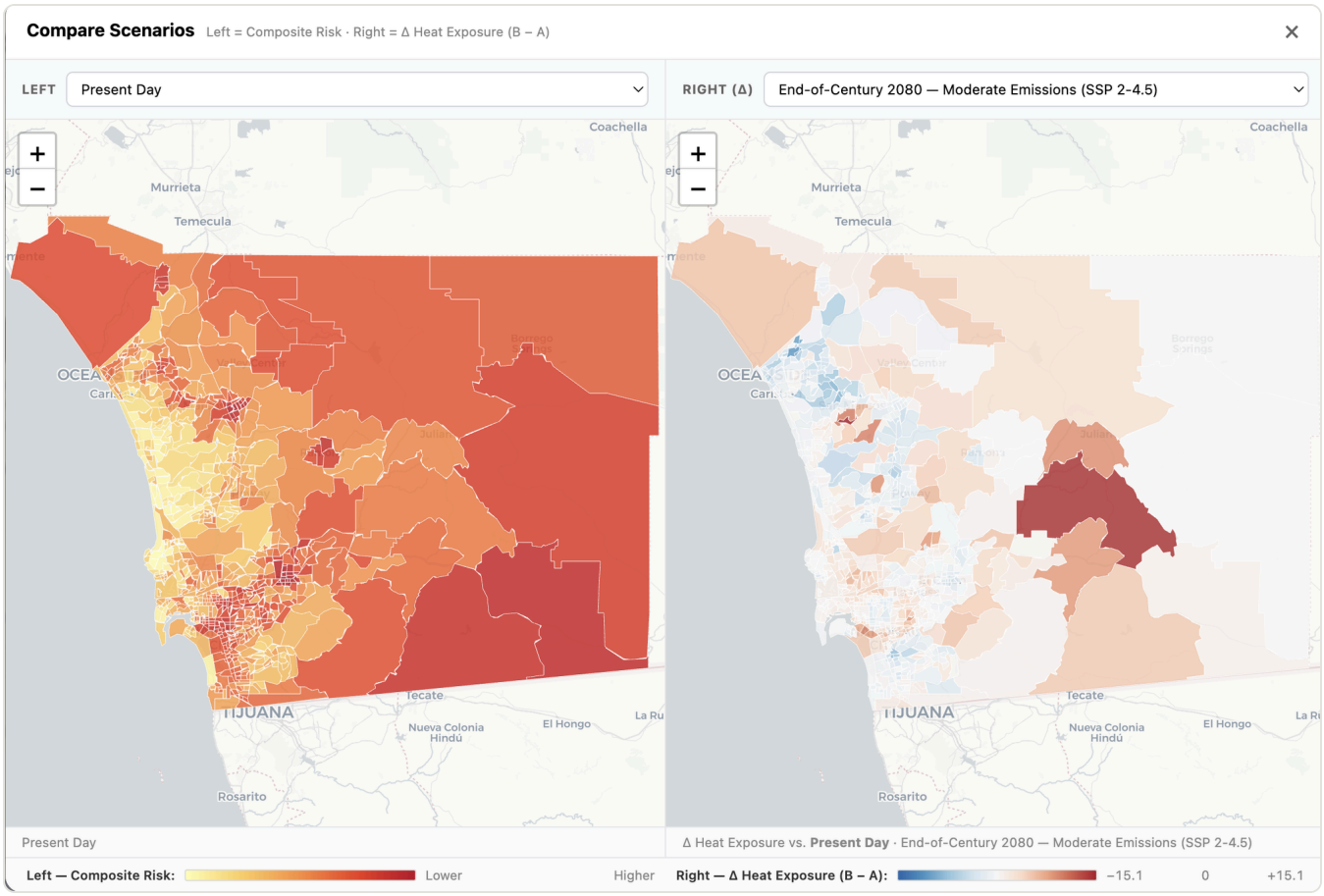
- **Present Day:** observed weather-station data.
- **2050:** mid-century mean (2035–2064) from LOCA2-downscaled CMIP6 projections.
- **2080:** end-of-century mean (2065–2094) from LOCA2-downscaled CMIP6 projections.

Selecting 2050 or 2080 expands an emissions-pathway selector with three Shared Socioeconomic Pathways (SSPs; see Carbon Brief explainer):

- **Moderate Emissions (SSP 2-4.5):** emissions decline but not aggressively.
- **High Emissions (SSP 3-7.0):** emissions remain high.
- **Very High Emissions (SSP 5-8.5):** unchecked fossil-fuel use.

Compare Scenarios Side-by-Side (button below the emissions-pathway selector) opens a paired view: the Heat Risk Index map under one scenario on the left, and the change in Heat Exposure pillar rank between two chosen scenarios on the right (B minus A, in percentile points on the 0–100 pillar scale — *not* degrees of warming). The two scenarios on either side can be set independently.

Why percentile rank instead of degrees? Across San Diego County, LOCA2 projects roughly similar absolute warming for every tract (typically ~1–3 °C between scenarios). A Δ°C map would therefore look nearly uniform. The percentile-rank view amplifies the spatial signal that matters for planning: *which tracts climb the relative heat-risk ladder fastest*. To anchor the rank shift in physical units, hover over any tract on the right map — the tooltip shows that tract's underlying LOCA2 Δ mean Tmax and Δ mean Tmin in °C.



Compare Scenarios view: present-day Composite Risk (left) alongside the change in Heat Exposure pillar rank for End-of-Century 2080 under SSP 2-4.5, relative to Present Day (right).

Population Sensitivity and Adaptive Capacity variables are held at present-day values. Reliable long-range projections for demographic, health, and infrastructure conditions don't yet exist at the census-tract level, so the tool does not project future changes to these variables.

Examples

Each example narrows one pillar to a single question and leaves the other two at their defaults, so the Heat Risk Index still reflects who is hot, who is vulnerable, and who can cope.

Start from the defaults, not from everything. The tool opens with a deliberately small selection: *Overall SVI*, *Age 65+*, and *Heat-Day Hospitalization Rate Ratio* for Population Sensitivity; *AC Prevalence (LACE)*, *Tree Canopy*, and *Income* for Adaptive Capacity; and the *Daytime heat* mode for Heat Exposure. Turning on all ~60 variables is not recommended — it pushes each pillar's PCA toward a generic “everything correlates” pattern and washes out the spatial contrast that makes the map useful. Unless an example says otherwise, leave the untouched pillars as they are.

The viewer also walks through Example 1 for you: take the welcome tour, and its final step offers a **See worked example** button that performs each step automatically.

Example 1: Where are children most at risk due to heat stress?

1. Click the **Population Sensitivity** header checkbox to clear that pillar, then expand *Sociodemographics* and check only **Age 17 & Under**.
2. Leave **Heat Exposure** on **Daytime heat** and leave **Adaptive Capacity** at its defaults.
3. Click **Build Composite Risk Map**.
4. The Population Sensitivity score is now driven entirely by the population 17 and under. Census tracts that are high risk (darker) **and** whose Dominant Driver is Population Sensitivity are places where children are most at risk due to heat stress.

Example 2: Where would people benefit most from subsidies to purchase AC units?

1. Click the **Adaptive Capacity** header checkbox to clear that pillar, then expand *Air Conditioning (AC)* and check only **AC Prevalence (LACE)**.
2. Leave **Heat Exposure** and **Population Sensitivity** at their defaults.
3. Click **Build Composite Risk Map**.
4. The Adaptive Capacity score is now a pure ranking of how AC-poor each census tract is. The darkest census tracts are priority areas for an AC-subsidy program: hot, sensitive, and currently lacking AC.

Two AC estimates are available. **AC Prevalence (LACE)** is the Census Bureau's official tract-level estimate; **AC Prevalence (Romitti)** is a published modeled estimate. Re-running the example with the other source is a quick way to check whether a tract's ranking depends on which estimate you trust.

Example 3: Where is nighttime heat most dangerous?

1. Under **Heat Exposure**, select the **Nighttime heat** mode. This one click replaces the whole exposure score — the mode averages Mean Overnight Minimum Temperature, 90th Percentile Overnight Minimum Temperature, and No-Relief Night Fraction.
2. Leave **Population Sensitivity** and **Adaptive Capacity** at their defaults.
3. Click **Build Composite Risk Map**.
4. The Heat Exposure score is now a pure ranking of how warm a census tract's nights are. The darkest census tracts have severe overnight heat alongside vulnerable populations and limited capacity to cool down.

Expect this map to look quite different from the default. Nighttime heat peaks in dense urban and coastal tracts, whereas daytime heat peaks inland — the two are close to spatially uncorrelated across the county. Switching back to **Daytime heat** and rebuilding is the fastest way to see which communities are exposed to one but not the other.

Feedback & bug reports

To submit a correction, comment, or feature request, open an issue on the project GitHub, or email cjmack@ucsd.edu.

If you're reporting something that looks wrong on the map, the census tract ID (shown at the top of the tract popup) and which heat mode and variables you had switched on are the two things that make it quickest to track down.

Data sources

VARIABLE	SOURCE
HEAT EXPOSURE	
Air temperature (daytime & overnight)	MesoWest / Synoptic Data
Humid heat (Wet-Bulb Globe Temperature)	NOAA HRRR
Land surface temperature	USGS Landsat
Future temperature projections	LOCA2 downscaled CMIP6
POPULATION SENSITIVITY	
Chronic disease & social needs	CDC PLACES
Social vulnerability	CDC SVI 2022
Heat-related hospitalizations	CalHeatScore (CalEPA / OEHHA)
Energy burden	DOE LEAD Tool (2022)
Demographics, housing & occupation	ACS 5-year estimates
ADAPTIVE CAPACITY	
Income, education, healthcare, parks & tree canopy	Healthy Places Index (HPI)
Air conditioning prevalence	U.S. Census LACE 2023; Romitti et al. (modeled)
Impervious surface	NLCD (MRLC)
Green space & land use	SANDAG / SanGIS
REFERENCE MAP LAYERS	
Population density	EPA EnviroAtlas

VARIABLE	SOURCE
Excess heat-related ER visits & heat-risk thresholds	UCLA Heat Maps (from California HCAI emergency-department records)
Cool-zone cooling sites	San Diego County Cool Zones — County HHSA site list & SanGIS feature service (designated sites only)
GEOGRAPHY & DERIVED LAYERS	
Census tract & city boundaries (also the geometry in GeoJSON downloads)	U.S. Census Bureau TIGER/Line
Composite risk, pillar scores & percentile ranks	Computed by this project — CC BY 4.0

Each variable name in the tool has a small `(Source)` link that opens the original source.

Privacy

The map runs entirely in your browser. There is no server, no account, and no analytics or tracking — we do not log who you are, what you look at, or what you build. Everything you select stays on your device, and the data you download is generated locally.

Two things do leave your browser, and both go to third parties rather than to us:

- **Map tiles.** As you pan and zoom, the background map is fetched from Esri. They see which part of the map you are looking at.
- **Address search.** Anything you type into the address box is sent to OpenStreetMap's Nominatim geocoder to be turned into a location. **If you search your home address, that address is sent to OpenStreetMap.** If you would rather not do that, click the map instead of searching — clicking sends nothing.

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